

MIMO Disturbance Observer for the Mecanum Platform

Derivation, Algebra, and Implementation Plan

1 Why MIMO (motivation)

The body mass matrix has off-diagonal couplings from the COM offset (a_X, a_Y) , and the controller's equivalent-control expressions already invert an *augmented* mass matrix that includes the wheel rotational inertia reflected through the rolling kinematics. A scalar (SISO) per-axis disturbance observer uses only the diagonal element of that inverse, so the off-diagonal coupling between V_y and $\dot{\psi}$ (the ma_X, ma_Y terms) is pushed into the lumped disturbance on *both* axes simultaneously. With two simultaneous SISO observers, the shared coupling is double-counted.

The MIMO formulation builds one 3-component observer that estimates a vector disturbance $\boldsymbol{\delta} = [\delta_x, \delta_y, \delta_\psi]^\top$. The rigid-body coupling is captured exactly by the augmented mass matrix inverse, so $\boldsymbol{\delta}$ absorbs only the truly unmodelled effects (friction-model error, indirect-control lag through the wheel/slip path, roller ripple, OOD- χ drift).

2 Notation and the augmented mass matrix

State and signal vectors. Body-frame velocity vector

$$\mathbf{v} = \begin{bmatrix} V_x \\ V_y \\ \psi \end{bmatrix} \in \mathbb{R}^3, \quad \dot{\mathbf{v}} = \begin{bmatrix} \dot{V}_x \\ \dot{V}_y \\ \dot{\psi} \end{bmatrix}. \quad (1)$$

Two distinct wrench representations live in the code. The first is the *code wrench* $\mathbf{M}_c = [M_x, M_y, M_\psi]^\top$ exactly as it appears in `asmc_torques`; it maps to the four wheel torques through the $\frac{1}{4}(\cdots)$ distributor with the yaw component carrying a factor $L = R/(l+h)$. The second is the *physical wrench* $\mathbf{F}_p = [F_x, F_y, M_z]^\top$ in SI units (N, N, N·m). The two are related by

$$\mathbf{M}_c = \text{diag}(R, R, 1) \mathbf{F}_p, \quad \mathbf{F}_p = \text{diag}(R^{-1}, R^{-1}, 1) \mathbf{M}_c, \quad (2)$$

because each unit of M_x or M_y in the code is one unit of wheel torque, which produces $1/R$ of body force at the contact, while M_ψ already carries its L -scaling explicitly inside the wheel-torque distributor.

Augmented mass matrix. The controller's `Mx_eq`, `My_eq` and `M_psi_eq` are exactly the rigid-body inverse dynamics with the wheel inertia J_1 reflected to the body through the rolling kinematics. Reading off the coefficients of $\dot{\mathbf{v}}$, the augmented mass matrix (symmetric, physical units) is

$$\mathbf{M}_{\text{aug}} = \begin{bmatrix} \tilde{m} & 0 & -ma_Y \\ 0 & \tilde{m} & ma_X \\ -ma_Y & ma_X & I_\psi \end{bmatrix}, \quad \tilde{m} \equiv m_s + \frac{4J_1}{R^2}, \quad I_\psi \equiv I_s + \frac{4(l+h)^2 J_1}{R^2}. \quad (3)$$

The off-diagonals $\pm m a_X$, $\pm m a_Y$ come from the COM offset and are unchanged by the wheel reflection (the wheels rotate about their own axes, which adds only to the diagonals).

Comparison with `PlatformParams.M_inv` in the existing code is the inverse of the *plain* body mass matrix used in the plant's RHS (no wheel reflection, because the wheel rotational dynamics are integrated separately as the four states ω_i). That is *not* the matrix we want here. The DOB must be consistent with the controller's view of authority, so it uses $\mathbf{M}_{\text{aug}}^{-1}$, not the plant's $\mathbf{M}^{-1}$. We will precompute $\mathbf{M}_{\text{aug}}^{-1}$ once at construction time.

3 Nominal plant model

We commit to a *nominal* input–output relation by which the controller and the observer agree on what authority the wrench has on the acceleration:

$$\dot{\mathbf{v}} = \mathbf{M}_{\text{aug}}^{-1} \mathbf{F}_p + \boldsymbol{\delta} \quad (4)$$

Equation (4) is a *definition* of $\boldsymbol{\delta}$: whatever isn't $\mathbf{M}_{\text{aug}}^{-1} \mathbf{F}_p$ is $\boldsymbol{\delta}$ by construction. The rigid–body coupling between the axes, including the wheel reflection, sits inside $\mathbf{M}_{\text{aug}}^{-1} \mathbf{F}_p$ and *does not* appear in $\boldsymbol{\delta}$. What *does* appear in $\boldsymbol{\delta}$: the friction–model residual (Coriolis terms the equivalent control did not fully cancel, viscous drag mismatch, slip–spin coupling), the indirect–control lag between commanding a wrench and the friction forces actually delivering that wrench through the wheels, the roller ripple, and any OOD parameter drift.

4 Observer structure

Choose a reduced–order observer with one auxiliary state per axis. Let $\boldsymbol{\zeta} \in \mathbb{R}^3$ be the auxiliary state and define the estimate by

$$\hat{\boldsymbol{\delta}} = \boldsymbol{\zeta} + \omega_0 \mathbf{v}, \quad (5)$$

where ω_0 is a *single scalar* observer bandwidth, the same for all three channels. The observer gain matrix is therefore $\boldsymbol{\Omega} = \omega_0 \mathbf{I}_3$. Using a scalar gain (rather than a diagonal matrix of three different ω 's) is the choice the user asked for: it keeps the algebra clean because $\boldsymbol{\Omega}$ commutes with every matrix in sight, and it gives one tuning knob instead of three.

5 Derivation of the auxiliary–state ODE

We want the estimate to track $\boldsymbol{\delta}$ through a first–order low–pass with cutoff ω_0 :

$$\dot{\hat{\boldsymbol{\delta}}} = -\omega_0 (\hat{\boldsymbol{\delta}} - \boldsymbol{\delta}). \quad (6)$$

This is the standard reduced–order observer error dynamics, lifted to three channels through the scalar ω_0 . Differentiating (5) and substituting the nominal model (4):

$$\dot{\hat{\boldsymbol{\delta}}} = \dot{\boldsymbol{\zeta}} + \omega_0 \dot{\mathbf{v}} = \dot{\boldsymbol{\zeta}} + \omega_0 (\mathbf{M}_{\text{aug}}^{-1} \mathbf{F}_p + \boldsymbol{\delta}). \quad (7)$$

Setting (7) equal to the target (6):

$$\dot{\zeta} + \omega_0 \mathbf{M}_{\text{aug}}^{-1} \mathbf{F}_p + \omega_0 \hat{\delta} = -\omega_0 \hat{\delta} + \omega_0 \delta. \quad (8)$$

The $\omega_0 \delta$ on each side cancels, which is the whole point of the construction — it lets the auxiliary state’s ODE be computed from *known* quantities only:

$$\dot{\zeta} = -\omega_0 \hat{\delta} - \omega_0 \mathbf{M}_{\text{aug}}^{-1} \mathbf{F}_p. \quad (9)$$

Finally substitute $\hat{\delta} = \zeta + \omega_0 \mathbf{v}$:

$$\boxed{\dot{\zeta} = -\omega_0 \zeta - \omega_0^2 \mathbf{v} - \omega_0 \mathbf{M}_{\text{aug}}^{-1} \mathbf{F}_p} \quad (10)$$

This is the ODE for the new 3-component state. The three terms have clean interpretations: $-\omega_0 \zeta$ is the natural low-pass relaxation, $-\omega_0^2 \mathbf{v}$ is the velocity contribution that pairs with the $+\omega_0 \mathbf{v}$ inside the estimate to make the algebra cancel, and $-\omega_0 \mathbf{M}_{\text{aug}}^{-1} \mathbf{F}_p$ is the wrench feedthrough that cancels the model’s $+b\mathbf{M}$ feedthrough.

6 Wrench recovery from the wheel torques

The auxiliary-state ODE (10) needs the *applied* physical wrench $\mathbf{F}_p$. The applied quantity in the plant is the four saturated wheel torques $M_i^{\text{sat}} = M_{\text{max}} \tanh(M_i^{\text{total}}/M_{\text{max}})$. Inverting the wrench-to-torque map gives the code wrench:

$$M_x^{\text{applied}} = M_1^{\text{sat}} + M_2^{\text{sat}} + M_3^{\text{sat}} + M_4^{\text{sat}}, \quad (11)$$

$$M_y^{\text{applied}} = -M_1^{\text{sat}} + M_2^{\text{sat}} + M_3^{\text{sat}} - M_4^{\text{sat}}, \quad (12)$$

$$M_\psi^{\text{applied}} = \frac{-M_1^{\text{sat}} + M_2^{\text{sat}} - M_3^{\text{sat}} + M_4^{\text{sat}}}{L}, \quad L = \frac{R}{l+h}. \quad (13)$$

Then convert to physical units via (2):

$$\mathbf{F}_p^{\text{applied}} = \begin{bmatrix} F_x^{\text{applied}} \\ F_y^{\text{applied}} \\ M_z^{\text{applied}} \end{bmatrix} = \begin{bmatrix} M_x^{\text{applied}}/R \\ M_y^{\text{applied}}/R \\ M_\psi^{\text{applied}} \end{bmatrix}. \quad (14)$$

7 Compensation wrench

The objective is for the closed-loop acceleration to track the nominal command without the disturbance term, i.e. to drive the net $\dot{\mathbf{v}}$ to $\mathbf{M}_{\text{aug}}^{-1} \mathbf{F}_p^{\text{nominal}}$. We add a corrective physical wrench $\Delta \mathbf{F}_p$ to the equivalent-control wrench such that, under the nominal model,

$$\mathbf{M}_{\text{aug}}^{-1} \Delta \mathbf{F}_p + \hat{\delta} = \mathbf{0}, \quad (15)$$

which immediately gives

$$\boxed{\Delta \mathbf{F}_p = -\mathbf{M}_{\text{aug}} \hat{\delta}} \quad (16)$$

and, in code wrench units,

$$\boxed{\Delta \mathbf{M}_c = -\text{diag}(R, R, 1) \mathbf{M}_{\text{aug}} \hat{\delta}} \quad (17)$$

The compensation is applied as $\mathbf{M}_c^{\text{eq}} \leftarrow \mathbf{M}_c^{\text{eq}} + \Delta\mathbf{M}_c$, i.e. subtracted from the equivalent-control wrench.

Writing (17) component by component, using (3):

$$\Delta M_x = -R(\tilde{m}\hat{\delta}_x + 0\hat{\delta}_y - m a_Y \hat{\delta}_\psi) = -R\tilde{m}\hat{\delta}_x + R m a_Y \hat{\delta}_\psi, \quad (18)$$

$$\Delta M_y = -R(0\hat{\delta}_x + \tilde{m}\hat{\delta}_y + m a_X \hat{\delta}_\psi) = -R\tilde{m}\hat{\delta}_y - R m a_X \hat{\delta}_\psi, \quad (19)$$

$$\Delta M_\psi = -1 \cdot (-m a_Y \hat{\delta}_x + m a_X \hat{\delta}_y + I_\psi \hat{\delta}_\psi) = m a_Y \hat{\delta}_x - m a_X \hat{\delta}_y - I_\psi \hat{\delta}_\psi. \quad (20)$$

Compared with the SISO version's $\Delta M_y = -R\tilde{m}\hat{\delta}_y$ and $\Delta M_\psi = -I_\psi \hat{\delta}_\psi$, the new cross-axis terms are the $\pm R m a_X \hat{\delta}_\psi$ in ΔM_y and the $-m a_X \hat{\delta}_y$ in ΔM_ψ — they say: “some of the $\dot{\psi}$ -axis disturbance has to be paid by a small M_y correction (and vice versa) because the rigid-body inertia matrix couples them.”

8 Error dynamics in terms of $\mathbf{e} = \hat{\boldsymbol{\delta}} - \boldsymbol{\delta}$

Define the estimation error

$$\mathbf{e} \equiv \hat{\boldsymbol{\delta}} - \boldsymbol{\delta} \in \mathbb{R}^3. \quad (21)$$

First evaluate $\dot{\hat{\boldsymbol{\delta}}}$ using the constructed $\dot{\boldsymbol{\zeta}}$ in (10) and the nominal model (4):

$$\begin{aligned} \dot{\hat{\boldsymbol{\delta}}} &= \dot{\boldsymbol{\zeta}} + \omega_0 \dot{\mathbf{v}} \\ &= (-\omega_0 \boldsymbol{\zeta} - \omega_0^2 \mathbf{v} - \omega_0 \mathbf{M}_{\text{aug}}^{-1} \mathbf{F}_p) + \omega_0 (\mathbf{M}_{\text{aug}}^{-1} \mathbf{F}_p + \boldsymbol{\delta}) \\ &= -\omega_0 (\boldsymbol{\zeta} + \omega_0 \mathbf{v}) + \omega_0 \boldsymbol{\delta} \\ &= -\omega_0 \hat{\boldsymbol{\delta}} + \omega_0 \boldsymbol{\delta} = -\omega_0 \mathbf{e}. \end{aligned} \quad (22)$$

Differentiating the error definition (21) and substituting (22):

$$\dot{\mathbf{e}} = \dot{\hat{\boldsymbol{\delta}}} - \dot{\boldsymbol{\delta}} = -\omega_0 \mathbf{e} - \dot{\boldsymbol{\delta}}. \quad (23)$$

$$\boxed{\dot{\mathbf{e}} = -\omega_0 \mathbf{e} - \dot{\boldsymbol{\delta}}} \quad (24)$$

The error dynamics are a stable first-order vector system at rate ω_0 , decoupled across the three channels (because $\omega_0 \mathbf{I}$ is diagonal), forced only by the time derivative of the true disturbance.

Two regimes.

- *Constant disturbance*, $\dot{\boldsymbol{\delta}} = \mathbf{0}$. The error decays exponentially,

$$\mathbf{e}(t) = \mathbf{e}(0) e^{-\omega_0 t}, \quad (25)$$

with time constant $1/\omega_0$ on each channel.

- *Time-varying disturbance*. The componentwise steady-state error is bounded by

$$|e_i|_{\text{ss}} \leq \frac{\sup_t |\dot{\delta}_i(t)|}{\omega_0}, \quad i \in \{x, y, \psi\}. \quad (26)$$

High-frequency content in $\boldsymbol{\delta}$ above ω_0 is *not* rejected — the transfer function from $\boldsymbol{\delta}$ to $\mathbf{e}$ peaks near ω_0 , which is the standard reduced-order DOB sensitivity waterbed.

Cancellation requirement (failure mode of the construction). The reduction of $\hat{\delta}$ to $-\omega_0 \mathbf{e}$ in (22) relies on the *same* $\mathbf{M}_{\text{aug}}^{-1}$ in the nominal model (4) and the observer ODE (10). If the two disagree, the residual $-\omega_0(\mathbf{M}_{\text{aug,model}}^{-1} - \mathbf{M}_{\text{aug,obs}}^{-1})\mathbf{F}_p$ stays in $\dot{\mathbf{e}}$ and the error is driven by the control signal directly — the observer becomes a control-driven amplifier rather than a disturbance tracker. The same $\mathbf{M}_{\text{aug}}$ (not its inverse) must appear in the compensation (16), or you carry a constant-gain steady-state bias.

9 Summary of changes relative to the SISO version

1. **State vector grows from 34 to 36.** Three auxiliary states $(\zeta_x, \zeta_y, \zeta_\psi)$ replace the one ζ used by the yaw-only SISO observer. The y-axis SISO state added previously is also folded in.
2. **Single observer bandwidth ω_0 .** The three per-axis ω 's collapse to one. `ESOParams` becomes `omega_o::Float64` and a single `enable::Bool`.
3. **Augmented mass matrix is precomputed.** Add $\mathbf{M}_{\text{aug}}$ and $\mathbf{M}_{\text{aug}}^{-1}$ as fields of `PlatformParams` (the existing `M_inv` stays untouched; that one is the plant's body matrix without wheel reflection and is used by the plant RHS).
4. **Observer ODE** in the plant RHS becomes the vector statement (10). `F_phys_applied` is recovered from the saturated wheel torques via (11)–(13) and (14).
5. **Compensation in `asmc_torques`** becomes the vector statement (17), or equivalently the per-axis (18)–(20). Note the new cross-axis contributions to ΔM_y and ΔM_ψ , and a fresh ΔM_x (which the SISO version did not have).
6. **Tracking property.** The error $\mathbf{e} = \hat{\delta} - \delta$ obeys (24); each component is a first-order channel at rate ω_0 driven by $\dot{\delta}_i$. The diagnostic overlay test from before generalises naturally: $\delta_i^{\text{true}} = \dot{v}_i - [\mathbf{M}_{\text{aug}}^{-1}\mathbf{F}_p^{\text{applied}}]_i$ vs $\hat{\delta}_i$, for $i \in \{x, y, \psi\}$.

A LaTeX source, one Word equation per block

Each `Verbatim` block below holds exactly one equation. Paste it into Microsoft Word's equation editor in LaTeX input mode. Compound statements from the main text are split into separate blocks so every entry is independently renderable. Standard math commands (`\frac`, `\mathbf`, `\hat`, `\dot`, Greek letters, subscripts/superscripts, `\bmatrix`) render correctly. `\boxed` is not Word-native; paste the inner expression without the wrapping.

From §2 – notation and the augmented mass matrix

Body velocity vector, eq. (1):

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\mathbf{v} = \begin{bmatrix} V_x \ \ V_y \ \ \dot{\psi} \end{bmatrix}
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Body acceleration vector, eq. (1):

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\dot{\mathbf{v}} = \begin{bmatrix} \dot{V}_x \ \ \dot{V}_y \ \ \ddot{\psi} \end{bmatrix}
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Code wrench in terms of physical wrench, eq. (2):

$$\mathbf{M}_c = \mathrm{diag}(R, R, 1), \mathbf{F}_p$$

Physical wrench in terms of code wrench, eq. (2):

$$\mathbf{F}_p = \mathrm{diag}(R^{-1}, R^{-1}, 1), \mathbf{M}_c$$

Augmented mass matrix, eq. (3):

$$\mathbf{M}_{\mathrm{aug}} = \begin{bmatrix} \tilde{m} & 0 & -m_{a_Y} \\ 0 & \tilde{m} & m_{a_X} \\ -m_{a_Y} & m_{a_X} & I_{\psi} \end{bmatrix}$$

Effective translational mass, eq. (3):

$$\tilde{m} = m_s + \frac{4 J_1}{R^2}$$

Effective yaw inertia, eq. (3):

$$I_{\psi} = I_s + \frac{4(1+h)^2 J_1}{R^2}$$

From §3 – nominal plant model

Nominal plant model, eq. (4):

$$\dot{\mathbf{v}} = \mathbf{M}_{\mathrm{aug}}^{-1} \mathbf{F}_p + \boldsymbol{\delta}$$

From §4 – observer structure

Observer estimate, eq. (5):

$$\hat{\boldsymbol{\delta}} = \boldsymbol{\zeta} + \omega_0 \mathbf{v}$$

From §5 – derivation of $\dot{\zeta}$

Target error dynamics for the estimate, eq. (6):

$$\dot{\hat{\boldsymbol{\delta}}} = -\omega_0 \hat{\boldsymbol{\delta}} - \boldsymbol{\delta}$$

Differentiating the estimate, eq. (7, step 1):

$$\dot{\hat{\boldsymbol{\delta}}} = \dot{\boldsymbol{\zeta}} + \omega_0 \dot{\mathbf{v}}$$

Substitute the nominal model, eq. (7, step 2):

$$\dot{\hat{\boldsymbol{\delta}}} = \dot{\boldsymbol{\zeta}} + \omega_0 \mathbf{M}_{\mathrm{aug}}^{-1} \mathbf{F}_p + \omega_0 \boldsymbol{\delta}$$

Equate to the target (LHS), eq. (8):

$$\dot{\boldsymbol{\zeta}} + \omega_0 \mathbf{M}_{\mathrm{aug}}^{-1} \mathbf{F}_p + \omega_0 \boldsymbol{\delta} = -\omega_0 \hat{\boldsymbol{\delta}}$$

After the $\omega_0 \boldsymbol{\delta}$ cancellation, eq. (9):

$$\dot{\boldsymbol{\zeta}} = -\omega_0 \hat{\boldsymbol{\delta}} - \omega_0 \mathbf{M}_{\mathrm{aug}}^{-1} \mathbf{F}_p$$

Final auxiliary-state ODE, eq. (10):

$$\dot{\boldsymbol{\zeta}} = -\omega_0 \boldsymbol{\zeta} - \omega_0^2 \mathbf{v} - \omega_0 \mathbf{M}_{\mathrm{aug}}^{-1} \mathbf{F}_p$$

From §6 – wrench recovery from wheel torques

M_x *un-map*, eq. (11):

$$M_x^{\mathrm{applied}} = M_1^{\mathrm{sat}} + M_2^{\mathrm{sat}} + M_3^{\mathrm{sat}} + M_4^{\mathrm{sat}}$$

M_y *un-map*, eq. (12):

$$M_y^{\mathrm{applied}} = -M_1^{\mathrm{sat}} + M_2^{\mathrm{sat}} + M_3^{\mathrm{sat}} - M_4^{\mathrm{sat}}$$

M_ψ *un-map*, eq. (13):

$$M_\psi^{\mathrm{applied}} = \frac{-M_1^{\mathrm{sat}} + M_2^{\mathrm{sat}} - M_3^{\mathrm{sat}} + M_4^{\mathrm{sat}}}{L}$$

Lever ratio:

$$L = \frac{R}{1+h}$$

Code→physical wrench conversion (applied), eq. (14):

$$\mathbf{F}_p^{\mathrm{applied}} = \begin{bmatrix} M_x^{\mathrm{applied}}/R \\ M_y^{\mathrm{applied}}/R \\ M_\psi^{\mathrm{applied}} \end{bmatrix}$$

From §7 – compensation wrench

Compensation requirement (cancel $\hat{\delta}$), eq. (15):

$$\mathbf{M}_a^{-1} \Delta \mathbf{F}_p + \hat{\delta} = \mathbf{0}$$

Compensation in physical wrench units, eq. (16):

$$\Delta \mathbf{F}_p = -\mathbf{M}_a \hat{\delta}$$

Compensation in code wrench units, eq. (17):

$$\Delta \mathbf{M}_c = -\mathrm{diag}(R, R, 1) \mathbf{M}_a \hat{\delta}$$

Per-axis compensation, x direction, eq. (18):

$$\Delta M_x = -R \tilde{m} \hat{\delta}_x + R m_{a_Y} \hat{\delta}_\psi$$

Per-axis compensation, y direction, eq. (19):

$$\Delta M_y = -R \tilde{m} \hat{\delta}_y - R m_{a_X} \hat{\delta}_\psi$$

Per-axis compensation, ψ direction, eq. (20):

$$\Delta M_\psi = m_{a_Y} \hat{\delta}_x - m_{a_X} \hat{\delta}_y - I_\psi \hat{\delta}_\psi$$

From §8 – error dynamics

Estimation error definition, eq. (21):

$$\mathbf{e} = \hat{\boldsymbol{\delta}} - \boldsymbol{\delta}$$

Estimate dynamics after substitution, eq. (22):

$$\dot{\hat{\boldsymbol{\delta}}} = -\omega_0 \hat{\boldsymbol{\delta}} + \omega_0 \boldsymbol{\delta}$$

Same identity in terms of the error, eq. (22):

$$\dot{\hat{\boldsymbol{\delta}}} = -\omega_0 \mathbf{e}$$

Error dynamics intermediate step, eq. (23):

$$\dot{\mathbf{e}} = \dot{\hat{\boldsymbol{\delta}}} - \dot{\boldsymbol{\delta}}$$

Error dynamics, final form, eq. (24):

$$\dot{\mathbf{e}} = -\omega_0 \mathbf{e} - \dot{\boldsymbol{\delta}}$$

Constant–disturbance solution, eq. (25):

$$\mathbf{e}(t) = \mathbf{e}(0)e^{-\omega_0 t}$$

Steady–state error bound, eq. (26):

$$\|\mathbf{e}\|_{\infty} \leq \frac{\sup_t \|\dot{\boldsymbol{\delta}}(t)\|}{\omega_0}$$

Notation glossary

$\mathbf{v}$	Body–frame velocity vector $[V_x, V_y, \dot{\psi}]^\top$.
$\mathbf{F}_p$	Physical wrench $[F_x, F_y, M_z]^\top$ in (N, N, N·m).
$\mathbf{M}_c$	Code wrench $[M_x, M_y, M_\psi]^\top$ as in <code>asmc_torques</code> .
$\mathbf{M}_{\text{aug}}$	Augmented mass matrix (with wheel reflection), see (3).
$\tilde{m}$	$m_s + 4J_1/R^2$, effective translational mass.
I_ψ	$I_s + 4(l+h)^2 J_1/R^2$, effective yaw inertia.
L	$R/(l+h)$, lever ratio in the wrench–to–torque map.
$\boldsymbol{\delta}$	Lumped disturbance in acceleration units; defined by (4).
$\hat{\boldsymbol{\delta}}$	Observer estimate of $\boldsymbol{\delta}$.
$\mathbf{e}$	Estimation error $\hat{\boldsymbol{\delta}} - \boldsymbol{\delta}$.
$\boldsymbol{\zeta}$	Auxiliary observer state (the integrator), three components.
ω_0	Single scalar observer bandwidth; $\boldsymbol{\Omega} = \omega_0 \mathbf{I}_3$.